Tools Documentation

TECHNICAL REBUILD REFERENCE
ARCHITECTURE, APIS, DATA FLOWS

TECHNICAL REFERENCE

How Each Tool Works

Complete documentation of every tool's architecture, API calls, system prompts, and data flows — everything needed to rebuild each tool in a new environment.

Platform Architecture

ZipJeweler runs as a **Next.js 14 serverless application** with a vanilla HTML/JS frontend served via iframe. All AI tool logic lives in a single HTML file. The backend provides proxy API routes that keep API keys server-side.

CORE API HELPER

All Claude-powered tools share a single `claudeAPI(body)` function that routes requests either through the server proxy or directly to Anthropic:

```
// Shared API wrapper — used by every Claude-powered tool
async function claudeAPI(body) {
  if (useServerKeys) {
    // Route through server proxy (keeps API key off client)
    return fetch('/api/chat', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(body)
    });
  }
  // Direct browser call (requires user-provided key)
  return fetch('https://api.anthropic.com/v1/messages', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'x-api-key': anthropicKey,
      'anthropic-version': '2023-06-01',
      'anthropic-dangerous-direct-browser-access': 'true'
    },
    body: JSON.stringify(body)
  });
}
```

BACKEND API ROUTES

POST /api/chat

POST /api/generate-image

Anthropic Messages API proxy. Forwards model, messages, system, max_tokens. Keeps ANTHROPIC_API_KEY server-side.

OpenAI Responses API (gpt-4.1). Image editing/generation with multi-turn corrections via previous_response_id. Used by C2R and S2I.

POST /api/generate-render

OpenAI Responses API (primary) + Higgsfield FLUX (fallback). Supports 1-4 input images, mask-based editing, aspect ratio control.

GET /api/config

Reports which API keys are configured (hasAnthropicKey, hasOpenAI, hasHiggsfield). Never exposes actual key values.

💬 Imagine — AI Project Brief Builder

Conversational AI that extracts 26 structured fields from natural jewelry design discussion ▶

🖼 Sketch to Image (S2I)

Converts hand-drawn jewelry sketches into photorealistic renders via OpenAI image editing ▶

🖥 CAD to Render (C2R)

Converts CAD wireframe screenshots into photorealistic product photos via OpenAI image editing ▶

📣 Marketing Agent

Generates Instagram captions, product descriptions, and e-commerce listings from product photos and specs ▶

$ Quick Estimate — Pricing Engine

Calculates Good/Better/Best pricing tiers using metal spot prices, stone costs, labor hours, and margin targets ▶

📐 Photo 3D (P3D) — 3D Prototype Planner

Generates CAD build specifications from jewelry reference photos (Building) ▶

🤖 Dashboard Assistant

Natural language interface for studio management — editing folders, uploading files, creating documents (Building) ▶

✓ Fidelity Grading (Shared)

Automated render quality scoring used by S2I and C2R to compare source vs. output ▶

⚙ Backend API Routes — Full Reference

Complete request/response schemas for all server-side proxy endpoints ▶